Jason Khatkar

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Mechanical Engineer | Design & Manufacturing | Portfolio

EDUCATION

University of California, Berkeley **August 2022-December 2024**
Bachelor of Science in Mechanical Engineering **GPA: 3.71/4.0**

Relevant Coursework:, 3D Modeling, Statics & Strength of Material, Manufacturing & Tolerancing, Thermodynamics, Material Mechanical Behavior, Manufacturing and Design Communication, Heat Transfer, Feedback Control Systems, Fluid Mechanics, Mechatronics, Dynamic Flight Controls, GD&T, FEA

SKILLS

Software: SolidWorks, AutoCAD, Fusion360, MATLAB, ANSYS
Programming & Tools: MATLAB, Arduino, Python, C++, Eclipse IDE, Overleaf, 3DOS, Microsoft 365,
Fabrication: Lathe, Mill, Waterjet, Bandsaw, Belt Sander, SLA & FDM Printing, Laser Cutting, Bandsaw
Certification: Engineer in Training (EIT), Machine Shop Certified

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RELEVANT EXPERIENCE

Crain Walnut Shelling: Mechanical Design Engineer **December 2025 – March 2026**

- Led redesign of a production line to integrate new machinery, including full layout updates, catwalks, handrails, and structural supports to reduce vibration, improve cleanability, conceal electrical systems, and meet OSHA standards
- Designed and implemented a cablevey product delivery system end-to-end, including chutes, grates, covers, inlets, and support structures to minimize wear and improve system longevity
- Created detailed CAD models, drawings, and facility layouts (SolidWorks, AutoCAD) and worked directly with welders and vendors to manufacture and install components to spec
- Collaborated with operators and specialists to develop custom tools and process improvements while managing 5–6 concurrent projects and researching industry-standard, food-safe manufacturing practices

CALSOL: Steering / Stability Lead **August 2023 – December 2024**

- Designed, fabricated, and tested of an array tilting mechanism to hold our delicate solar panel array at a range of angles
- Performed simulations and calculations driving design of shell, battery, and mechanical components to accommodate a 50% increase in shell without excessive weight
- Sourced parts, consulted catalogs, and CAD designed functioning parts for a new generation of steering, resulting in 40% more weight efficient through asymmetric design
- Documented and communicated design decisions across subteams supporting manufacturing, feasibility analysis, and sourcing

Space Technology and Rocketry: Lead Testing Engineer **August 2022 – May 2023**

- Developed complex CAD models, lathe, and waterjet to synthesize a test stand being used for pyro and cryo flow rates and force calculation in the prototype liquid engine of our rocket
- Assembly, disassembly, and physical creation of the valve configuration as well as handling fuel and cryo-fluids
- Collaborated in testing and handling of UC Berkeley's first liquid propulsion engine for the Lander Challenge

RELEVANT PROJECTS

ME 102B (Mechatronics Design): Automated Pill Dispenser **Fall 2024**
Class Project: Create a device to support the elderly

- Waterjet, 3D printed, milled, and reamed parts to contribute to a working prototype
- Wrote professional reports and a BOM to explain and justify product design choices
- Iteratively improved CAD designs to resolve internal stress during assembly, ultimately winning us the Societal Impact Award

ME 130 (Design of Planar Machinery): Desk Lift **Fall 2023**

- Designed a motor-driven mechanism to move a keyboard tray through two toggle positions
- Performed motion and structural analysis to validate performance prior to manufacturing
- Used material properties and complex geometrical interfaces to generate graphs that validated design before manufacturing